

Maintenance, Inspection and Troubleshooting

Robot Weld 01 (robot-weld-01) | Sensor 1: AXIS_4_SERVO_TORQUE

Machine	Robot Weld 01 (machine ID robot-weld-01, six-axis spot-welding robot, Body shop, site vit-lab)
Sensor	Sensor 1: AXIS_4_SERVO_TORQUE, "Axis 4 servo torque", sensor ID ROBOT-WELD-01.AXIS_4_SERVO_TORQUE, type torque, unit Nm
Component / subsystem	Robot forearm (axis 4): axis 4 servo drive and gearbox; planned agent component tag robot_forearm_axis_4; suggested twin mesh 105-six-axis-welding-robot/arm-swing
Document title	Maintenance, Inspection and Troubleshooting
Document ID	RW01-S1-DOC-04
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Purpose	Translates abnormal Sensor 1 (axis 4 servo torque) conditions into practical inspection, troubleshooting, corrective-action, verification and return-to-service steps for the axis 4 servo drive and gearbox of Robot Weld 01.
Scope	Maintenance of the axis 4 drive train (robot forearm) and of the Sensor 1 data path. No repository runbook exists for the robot, so procedures are engineering-reference [E] and must be replaced or confirmed by the robot manufacturer's manual. Interpretation levels are in Documents 01 to 03; cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_SENSOR1.md (catalog limits, simulation equations, topics, tests).
[D]	Derived value: computed by the document authors from the repository's own simulation equations (noise-free unless stated).
[E]	Engineering reference: general industrial practice for robot servo-axis diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. An ESP32 gateway (or a software simulator) plays the robot controller and generates values from documented equations. The limits in this document that are tagged [P] are illustrative constants chosen for the demo in the repository, not OEM specifications. Values tagged [E] and [I] are not manufacturer-certified.

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1. Maintenance Philosophy

1.1 Condition-based approach

The repository provides no inspection intervals, lubricant specifications, backlash limits, spare-part numbers, torque values or mean time between failures for axis 4 [P]. This document therefore does not invent intervals. Maintenance is triggered by condition: the Sensor 1 level (L1 to L4, LS), the stage of the gear train (stages 0 to 5) and the confirmed diagnosis. Routine inspections follow the OEM maintenance schedule and the site plan, which are referenced here but not reproduced.

PRINCIPLE: compare the channel to its own baseline and look at the shape of the signal; do not run a drifting component to alarm; schedule inspection at the next planned stop (carried over from the repository's press runbook stubs, applied here by analogy [P][E]).

Table 1. Condition-based maintenance triggers

Sensor 1 condition	Level / stage	Maintenance response	Timing basis
Healthy: M 12.5 to 13.5 Nm, A6 at most 0.2 Nm	L0 / stage 0	Routine inspections per OEM schedule	OEM schedule (not specified in repository)
M 13.5 to 15.0 Nm or A6 0.4 to 0.6 Nm	L1 / stage 1	Add to next planned inspection; check lubrication records; trend	Next planned inspection
M 15.0 to 18.0 Nm, no warn	L1 to L2 / stage 2	Plan gearbox inspection (oil, backlash, fasteners, dress pack)	Within days (real) [I]
Peak at or above 26 Nm for 2 cycles	L2 / stage 3	Inspection at the next planned stop	Next planned stop
Peak at or above 30 Nm for 2 cycles	L3 / stage 4	Intervention in the shift; plan repair and parts	Within shift [I]
Sample at or above 34 Nm, or collision	L4 / stage 5	Immediate stop; full inspection	Immediate
Flat, clipped or noisy signal	LS	Data-path and signal inspection first	Within shift

1.2 What the repository does and does not provide

Table 2. Maintenance information availability

Topic	Status in repository	Treatment in this document
Axis 4 maintenance intervals, lubricant spec, backlash limit, torque values	Not specified in repository	Condition-based criteria; OEM values required [E]
Runbook pattern (front matter, scope, healthy signature, developing fault, why thresholds are not enough, telling faults apart, action, safety)	Available from the press stubs	Followed in structure
Safety wording: follow the manufacturer's procedure, lock-out and tag-out before opening the drive, not a replacement for the OEM manual	Available	Applied in Section 2
Agent constraint: cannot act on the machine; outputs are a report and user-interface commands	Available	Recommendations only
Data-path troubleshooting table	Available (firmware README, MQTT contract, demo script)	Reproduced in Section 12
Mechanical failure history for the robot	Not specified in repository	Cases in Document 05 are simulated scenarios

2. Safety Considerations

This document is a capstone prototype. It does not replace the robot manufacturer's manual, the plant's lock-out and tag-out procedure or site safety rules. Follow the manufacturer's procedure and apply lock-out and tag-out before opening the drive, gearbox or brake. Only qualified persons may perform the tasks below.

Table 3. Hazards and controls for axis 4 work

Hazard	Where	Control [E]
Gravity fall of the forearm when the brake is released or the drive is opened	Axis 4 and attached wrist, tool and gun	Support the arm mechanically or move it to the OEM service pose; never release the brake without support
Stored electrical energy in the drive	Servo drive, DC link	Lock-out and tag-out; wait for the capacitor discharge time stated by the OEM; verify zero energy
Unexpected motion	Whole robot	Lock-out the controller; enabling device in hand; cell guarding closed or fenced
Hot surfaces and lubricant	Gearbox, motor, brake	Allow cooling; use gloves; handle hot lubricant carefully
Pressure in the gearbox	Gearbox housing	Release pressure per OEM before opening plugs
Pinch points and crush	Forearm, tool, gun	Keep body outside the swept range
Weld gun and cabling	Spot weld gun and dress pack	Isolate welding power; discharge capacitors; remove tip pressure
Electrical shock	Motor and brake supply	Qualified electrician; use rated meters
Chemical exposure	Grease, oil, cleaners	Safety data sheets, gloves, eye protection
Restart after maintenance	Controller	Verification run at reduced speed first; clear the cell

2.1 Pre-work checklist

- Confirm the machine ID (robot-weld-01), axis (4) and the level from the agent report.
- Obtain the OEM manual, the maintenance permit and the lock-out and tag-out plan.
- Park the robot in the service pose, support the forearm and verify zero energy.
- Record the baseline readings before the work: Sensor 1 features, TCP mean, gun temperature, backlash if measurable.
- Prepare spare parts and lubricants identified from the OEM documentation, not from this document.

3. Preventive Inspection Checklist

The checklist is organised by component. The interval column states the basis, not a number, because intervals are OEM-specific. A telemetry trigger can advance any inspection.

Table 4. Preventive inspection checklist for the axis 4 drive train

Item	What to check	Interval basis	Telemetry trigger that advances it	Record
Gearbox lubricant	Level, colour, odour, metallic particles, leakage	OEM schedule	M above 13.5 Nm or A6 above 0.4 Nm	Level, condition, sample ID
Seals and breather	Weeping, cracks, blocked breather	OEM schedule	Noise rise	Condition
Fasteners (gearbox to arm, motor to gearbox)	Marks intact, torque check per OEM	OEM schedule; after any collision	Spikes at reversals	Torque values
Backlash at flange	Dial indicator measurement	OEM schedule; baseline at commissioning	dTCP above 0.15 mm	Value, direction, date
Brake	Release click, supply voltage, drag by hand	OEM schedule	Idle offset above 0.5 Nm	Voltage, timing
Motor and cable	Connector tightness, insulation, no chafing	OEM schedule	Offset or non-harmonic lines	Condition
Dress pack and hoses	Routing, slack, wear, spatter	Each planned stop	Phase-locked bump	Condition
Covers and spatter shields	Spatter build-up	Each planned stop	Slow rise of M	Cleaned
Gateway and data path	status online, synced true, no seq gaps, diag fields	Weekly for the prototype [I]	Stale or gaps	diag snapshot
Sensor 1 baseline	Recompute M, A6, sigma, template	After any repair or program change	Any L1	Baseline record
TCP accuracy	Mean deviation	After any collision or repair	dTCP rising	Mean in mm

4. Visual and Mechanical Inspection

4.1 Visual inspection

Table 5. Visual inspection table

Item	Look for	Finding	Meaning (link to Sensor 1)	Action
Gearbox housing	Oil weeping, stains, cracks	Wet seal, dark streaks	Lubrication loss (F02) leading to F01	Clean, find leak path, relubricate
Gearbox plug or magnetic plug	Metallic fines or flakes	Metal on magnet	Wear (F01, F04)	Take oil sample; plan gear inspection
Fasteners and torque marks	Marks broken or shifted	Misaligned marks	Looseness (F05)	Retorque and replace damaged fasteners
Dress pack	Rub marks, kinks	Tension at a pose	External drag (F06)	Re-dress
Brake area	Discolouration, dust	Hot marks	Brake drag (F03)	Electrical and mechanical brake check
Cables and connectors	Chafing, burn marks	Damaged insulation	Electrical fault (F08)	Replace and test
Tool and gun	Deformation, loose parts, collision marks	Damage	Overload or collision (F07)	Inspect the gear train
Cooling	Dust on fins, fan operation	Blocked	Thermal (F09)	Clean
Weld spatter	Build-up near seals	Spatter	Drag, seal damage	Clean and shield

4.2 Mechanical inspection

Table 6. Mechanical inspection table

Component / test	Method	Indicator of fault	Acceptance basis	Linked fault
Gearbox backlash	Dial indicator at the output flange, servo off and brake per OEM procedure	Backlash above OEM limit or increased by more than 50 percent from commissioning value [I]	OEM limit (not specified)	F05, F01
Output shaft play	Radial and axial movement by hand and indicator	Detectable play	OEM limit	F04
Noise at low speed	Slow axis move with all other axes still, listen with stethoscope	Roughness or periodic knock	Qualitative; compare with healthy axis	F01, F04
Oil or grease sample	Collect per OEM; send to laboratory or screen visually and magnetically	Metal flakes, colour change, increase of iron content versus previous sample [E]	OEM or laboratory limit	F01, F02
Temperature	Contact thermometer or IR on gearbox and motor housing at steady state	Higher than neighbouring axes or than previous record	OEM limit	F02, F03, F09
Torque to turn by hand (servo off, brake released with support)	Where permitted by OEM, feel resistance	Notchy or tight	Qualitative	F01, F03, F04
Fastener torque	Calibrated torque wrench	Below specification	OEM torque table	F05
Coupling or key	Inspect for fretting and play	Wear	OEM	F05
Mastering position	Compare mastering marks and readings	Offset after collision	OEM	F07

5. Electrical, Sensor and Wiring Inspection

5.1 Electrical inspection

Table 7. Electrical inspection table

Item	Test	Healthy	Fault indication	Linked fault
Brake supply	Voltage at brake terminals during release	Within OEM specification	Low voltage, late release	F03
Brake release timing	Controller log or scope	Release before motion	Late release	F03
Motor phase currents	Clamp meter or drive diagnostic	Balanced phases	Imbalance	F08
Current sensor offset	Drive diagnostic with motor idle	Zero current reads zero	Offset	F08
Encoder signal	Drive diagnostic	No errors	Noise or dropouts	F08, F04
Motor cable and connectors	Insulation test, visual	Within OEM	Low insulation, burn	F08
Grounding and shielding	Continuity	Within OEM	Poor ground	F08, F11
Drive temperature and faults	Drive log	Normal	Over-temperature or alarms	F08, F09

5.2 Sensor inspection: the Sensor 1 data path

Sensor 1 is a controller-internal value, not a separate transducer. Inspection therefore concerns the value path: controller estimate, gateway, broker and backend. In the prototype the ESP32 generates the value from an equation, so a data-path check is the first step when a reading looks implausible.

Table 8. Sensor validation table

Validation	Method	Expected result	Pass criterion	Failure means
Value follows the cycle	View 16 samples per cycle	8 s waveform, swing 7 Nm, mean 13 Nm in healthy state	Autocorrelation peak at 8 s	F11 or F10
Variance	Compute sigma at same phase	0.45 Nm (healthy)	0.40 to 0.55 Nm	Flat or burst signal
Range	Min and max	3.1 to 23.1 Nm (healthy)	Within range	Clipping, offset
Offset at idle	Torque with axis holding	Posture-dependent hold value	Matches previous record within 0.5 Nm	F08, F11, F03
Cross-check with independent current measurement	Clamp-on current at the motor, convert with torque constant (OEM)	Within about 5 percent after conversion [E][I]	Agreement	Controller estimate error
Known-load check	Move with and without a known load per OEM	Torque change proportional to load	Proportional	Gain error
Gateway health	diag fields uptimeS, rssi, heapFree, heapMin, reset, wifiReconnects, mqttReconnects, restarts, publishFailures, tls	Stable minimum heap over a 30 minute soak; no reconnects	No growth	Gateway fault (F12)
Sequence continuity	seq counter	No gaps (reboot shows new bootId with seq 0)	Gaps below 1 percent	Network loss
Clock	synced flag	true	true	NTP blocked
Cross-channel consistency	Compare torque with TCP and gun temperature	Moves as expected for scenario	Consistent	Identity or path error

5.3 Connection and wiring inspection

- Controller to gateway link: connector seating, cable condition and shield termination (real plant: OPC UA or MTConnect link and the gateway power supply) [E].
- Gateway power and Wi-Fi: stable supply, 2.4 GHz network, signal strength (rssi) not marginal; the prototype ESP32 runs on a USB or bench supply [P].
- Broker access: credentials with permission to publish on cdo/v1/vit-lab/robot-weld-01/...; TLS clock requirement [P].
- Motor power, brake and encoder cables of axis 4: connectors tight, no chafing in the dress pack, shield continuity [E].

6. Component Condition Assessment

Table 9. Component condition grades and their relation to Sensor 1 stages

Grade	Definition	Axis 4 gearbox evidence	Sensor 1 stage	Decision
A Good	No wear indications	Clean lubricant, backlash at baseline, no noise	Stage 0 (M 12.5 to 13.5 Nm, A6 at most 0.2)	Continue; routine
B Watch	Early indications	Slight discolouration, backlash increase below 25 percent [I]	Stage 1 to 2	Trend; plan inspection
C Degraded	Confirmed wear	Fines on magnet, backlash rise above 50 percent [I], noise	Stage 3	Repair at next stop
D Failed or failing	Damage affecting function	Chipped teeth, large play, heavy debris	Stage 4 to 5	Replace before return to service

The grades combine measurements and Sensor 1 evidence: a grade C component with Sensor 1 still in stage 2 shows that telemetry lags physical wear; the physical finding governs. Conversely stage 3 or higher in Sensor 1 with grade A physical findings points to a measurement or non-gearbox cause and requires rechecking the data path and the friction candidates.

6.1 Lubrication and service considerations

- Grade, quantity, fill procedure and replacement conditions come from the OEM manual; they are not specified in the repository [P].
- Never mix incompatible greases; record grade and batch at every service.

- After lubricant service, run warm-up cycles and recapture the Sensor 1 baseline; a new offset after service suggests over-fill or wrong grade.
- If metallic particles are found, the diagnosis moves from lubrication (F02) to wear (F01); a simple relubrication is no longer sufficient.
- Keep drains and breathers clean to avoid pressure and contamination; check seals whenever the lubricant is changed.

6.2 Alignment and installation checks

Table 10. Alignment and installation checks

Check	Method	Why it matters	Linked fault
Gearbox-to-arm mounting	Fastener torque marks; flatness of interface	Looseness gives reversal spikes	F05
Motor-to-gearbox coupling	Concentricity per OEM	Misalignment loads bearings	F04, F05
Mastering (zero position) of axis 4	Master per OEM after repair or collision	Position error appears as TCP deviation	F07, F01
Tool and payload data	Controller tool data matches real mass and centre of gravity	Wrong data gives wrong torque feedforward and payload faults	F10, F07
Dress pack routing	Slack across stroke	Drag bump	F06
Bracket and gun mounting	Torque and play	Added load or vibration	F07

6.3 Component inspection table

Table 11. Component inspection table for the axis 4 drive train

Component	Inspection method	Healthy finding	Fault finding	Corrective action	Linked Sensor 1 pattern
Axis 4 gearbox gear mesh	Oil and magnet check, noise, backlash	Clean lubricant, backlash at baseline	Flakes, chips, noise, backlash rise	Replace gear set or gearbox	Mean up, A6 above 0.4, noise up, TCP up
Gearbox lubricant	Level, colour, sample	Correct level, clear	Dark, milky, low	Replace or top up	Mean up, A6 absent
Gearbox and motor seals	Visual	Dry	Weeping	Replace seals	Slow mean rise
Output and motor bearings	Play, temperature, noise	No play, normal temperature	Play, hot, rumble	Replace bearings	Noise up, sparse spikes
Holding brake	Voltage, timing, drag, temperature	Releases fully	Drag, hot	Repair or replace	Mean up, idle offset
Servo drive and current sensors	Diagnostic log, balance	Balanced, no offset	Offset, imbalance	Replace or calibrate	Offset at idle, non-harmonic lines
Motor cable, encoder cable	Visual, continuity	Intact	Chafed, loose	Replace or reseal	Erratic noise, offset
Mounting and coupling	Torque, play	Marks aligned	Loose, fretting	Retorque or replace	Spikes at reversals, TCP up
Dress pack	Routing, slack	Free	Tight, worn	Re-dress	Bump at one phase
Tool and gun	Mass data, damage	Matches data	Damage or wrong data	Correct	Step, swing change
ESP32 gateway (data path)	diag fields	Stable heap, no reconnects	Reconnects, restarts	Fix network or firmware	Gaps, stale, flat

7. Calibration and Validation

Table 12. Calibration and validation activities

Activity	When	Procedure	Result recorded
Baseline recapture for Sensor 1	After repair, lubrication service, program change, payload change	Document 03 verification run: 20 cycles after warm-up	M, template, sigma, A6, peak
TCP verification	After repair or collision	Compare TOOL_CENTER_POINT_DEVIATION mean with healthy 0.104 mm and limit 0.25 mm [P]	Mean and max
Mastering check	After repair or collision	Per OEM	Mastering data
Torque estimate validation	When Sensor 1 is suspect	Independent current measurement or known-load test	Scale error percent
Limit review	When baseline changes	Recompute watch levels; keep platform limits 26 and 30 Nm unless changed in the catalog	Updated thresholds
Gateway validation	After firmware or catalog change	Self-test (fw selftest) compares C++ and JavaScript models to 12 significant digits [P]	Pass or fail

8. Troubleshooting Procedure

8.1 Procedure steps

Table 13. Troubleshooting procedure for an abnormal Sensor 1

Step	Action	Decision	Next
1	Read the agent report: level, pattern, diagnosis, confidence	Is the data path valid?	If no: Section 12
2	Confirm the abnormal reading on a second view (twin, history) and check operating state	Is state production and steady?	If no: do not diagnose; correct context
3	Check for recent events: collision, program or tool change, lubrication, repair	Event found?	Check F07, F10, F02 first
4	Perform the first inspection from the priority table	Finding confirms?	If yes: corrective action; if no: next
5	Perform the decisive test for the second candidate	Finding confirms?	As above
6	Implement corrective action in priority order	-	Section 9
7	Run verification (Section 10)	Criteria met?	If no: reopen and escalate
8	Record findings and update baseline	-	Close

8.2 Observed condition to action

Table 14. Troubleshooting table: observed Sensor 1 condition, cause, inspection, corrective action, verification

Observed Sensor 1 condition	Possible cause	Inspection	Corrective action	Verification
M 13.5 to 15.0 Nm, A6 0.4 to 0.6, no limit	Incipient gear wear or early friction	Review lubrication records; take oil sample at next stop	Trend; relubricate if due	M back within 12.5 to 13.5 and A6 at most 0.2 after service or stable trend
M 15 to 18 Nm, A6 0.6 to 1.5, TCP rising, status normal	Gearbox wear stage 2 (F01)	Oil, magnet, backlash, noise	Plan gear replacement; reduce duty	Features healthy after repair
Peak 26 to 30 Nm each cycle, A6 1.5 to 2.5	Gearbox wear stage 3	Same plus full gearbox inspection	Replace gear set or gearbox at next stop	Peak at most 23.1 Nm
Peak 30 Nm or above each cycle	Gearbox wear stage 4 or overload	Immediate inspection	Intervene; replace	Status normal for 600 s
Sample at or above 34 Nm	Collision or beyond-model wear	Stop; damage check	Specialist repair	Full verification
M high, A6 absent, idle offset	Brake drag or drive offset	Brake voltage, release timing; drive log	Repair brake or sensor	Idle offset zero
M high after cold start, falls with warm-up	Cold viscosity or lubricant degradation	Lubricant grade and condition; temperature	Service lubricant	Warm-up offset at most 0.5 Nm
M high, A6 absent, no idle offset, swing unchanged	Friction (F02), external drag (F06) or thermal	Lubricant, dress pack, temperature	Correct	M within band
M and swing both changed	Payload or program change	Tool data, program log	Correct tool data or re-baseline	New baseline
Spikes at reversals	Backlash or looseness	Fasteners, backlash, coupling	Retorque or replace	Spikes absent
Bump at one phase	Dress pack drag	Routing, slack	Re-dress	Bump absent
Noise rise with sparse spikes	Bearing degradation	Play, temperature, grease	Replace bearing	Noise 0.40 to 0.55 Nm
Flat constant signal	Signal fault, gateway freeze	diag fields, controller state	Restart gateway; fix path	Cycle returns
Values at 0 or 45 Nm	Saturation or scaling error	Scaling, drive	Fix scaling	Normal range
Torque normal but TCP rising	Backlash, mastering, other axis	Backlash, mastering	Correct	TCP at 0.10 mm
Torque warn but gun temperature high	Process or thermal event	Gun channel	Separate gun fault	Both normal
No data, device stale	Network or power	Gateway diag, power	Restore	Online
seq gaps	Packet loss	rsi, network	Improve link	No gaps
Alarm not cleared after repair	Alert latency or residual fault	Recheck features; 2-cycle clear rule	Reevaluate	Normal 600 s

9. Corrective Actions, Replacement Criteria and Priorities

Table 15. Fault-to-maintenance-action matrix

Fault	Primary action	Secondary action	Parts or consumables (OEM)	Skill level	Typical downtime [I]
F01 gearbox wear	Replace gear set or gearbox	Replace lubricant and seals; re-master; re-baseline	Gearbox or gear set, seals, lubricant	Specialist	Several hours to a shift
F02 lubrication	Relubricate or replace lubricant	Replace seals; clean breather	Lubricant, seals	Technician	1 to 2 hours
F03 brake drag	Repair or replace brake	Verify release timing	Brake assembly	Electrician and technician	1 to 3 hours
F04 bearing	Replace bearings	Relubricate; check shaft	Bearing set	Specialist	Several hours
F05 looseness	Retorque; replace coupling	Measure backlash	Fasteners, coupling	Technician	1 to 2 hours
F06 external drag	Re-dress, clean	Replace guides	Clamps, covers	Technician	30 to 60 min
F07 overload or collision	Inspect, repair damage, re-master	Check tool data	Depends	Specialist	Variable
F08 drive fault	Replace sensor or drive module	Recalibrate	Drive, sensor, cable	Electrician	1 to 3 hours
F09 thermal	Restore cooling; reduce duty	Allow warm-up	Filters	Technician	30 min
F10 program change	Re-baseline	Update tool data	None	Programmer	15 min
F11 sensor fault	Fix signal path	Restart gateway	Firmware or wiring	Controls engineer	30 to 60 min
F12 data path	Fix network, clock, identity	Restart	None	IT or controls	15 to 60 min

Table 16. Corrective-action priority table

Priority	Condition	Reason	Action first
1 Safety	L4, suspected collision, brake or drive fault	Risk to persons and equipment	Stop, lock-out, inspect
2 Damage prevention	L3 alarm, noise and temperature rising	Gearbox failure risk	Intervene in shift
3 Quality	L2 warning, TCP above 0.25 mm	Weld position quality	Plan inspection at next stop
4 Condition	L1 watch	Early detection	Plan, trend, check lubrication
5 Data	LS, F12	Blind spots	Fix path in parallel

9.1 Replacement criteria

Table 17. Replacement criteria

Component	Replace when [I]	Basis
Gear set / gearbox	Backlash above the OEM limit, chipped or pitted teeth, metal debris with rising trend, or Sensor 1 stage 4 confirmed by inspection	OEM limit plus inspection
Lubricant	Contaminated, discoloured or at OEM replacement condition; after any component replacement	OEM
Seals	Leakage, cracking, after opening the gearbox	OEM
Bearings	Detectable play, roughness, temperature rise confirmed	OEM
Brake	Drag confirmed, worn disc, supply voltage correct but release late	OEM
Coupling or key	Fretting or play	OEM
Drive or current sensor	Offset or imbalance confirmed and not correctable by calibration	OEM
Cables and dress pack	Damaged insulation, kinks, wear	OEM

10. Task Procedures

The task procedures below are engineering-reference outlines [E]. Quantities, tools, torque values, lubricant grades and sequences must be taken from the OEM manual. Each task ends with the data that must be recorded so that Sensor 1 features can later be compared with physical findings.

Table 18. Task T1: gearbox lubricant sampling and magnetic plug inspection

Step	Action	Check	Record
1	Obtain permit; lock-out; support forearm; verify zero energy	Permit and lock-out in place	Permit number
2	Clean around the sampling or drain point to avoid contamination	No debris near the opening	-
3	Open the plug or sampling port per OEM (release pressure first)	Pressure released	Lubricant state
4	Collect sample into a clean labelled container (about the volume the laboratory requests)	Sample from mid-flow, not from the sump bottom	Sample ID, axis, date, operating hours if known
5	Inspect magnetic plug: fines (fine grey paste) versus flakes or chips	Fines: normal run-in or slow wear; flakes or chips: damage	Photo, description
6	Check colour, odour, milkiness (water) and grit between fingers (gloved)	Dark, burnt odour, milky or gritty	Description
7	Refit plug with new seal per OEM; clean	No leak	Torque value used
8	Send sample to laboratory for wear metals and viscosity (limits per laboratory and OEM)	Report received	Results and date

Table 19. Interpretation of lubricant findings in relation to Sensor 1

Finding	Interpretation [E]	Sensor 1 link	Next step
Clean lubricant, no particles	Lubricant and gears in good condition	Mean shift without ripple points to friction, brake or sensor	Check brake, drive, temperature
Fines on magnet only	Normal run-in or slow wear	Stage 0 to 1 if M unchanged	Trend
Flakes, chips or glittering metal	Surface fatigue or tooth damage	Matches stage 2 to 4 with A6 present	Plan gear inspection and replacement
Dark, burnt or oxidised lubricant	Overheating or ageing	F02, F09; M up without ripple	Replace lubricant; find heat source
Milky lubricant	Water ingress	F02 with seal fault	Replace lubricant and seals
Wear metal level rising versus previous sample	Active wear	Trend confirms F01	Shorten sampling interval; plan repair

Table 20. Task T2: backlash measurement at the axis 4 flange

Step	Action	Check	Record
1	Lock-out; support forearm per OEM; use the OEM method for holding the motor side	Motor side held per OEM	-
2	Mount a dial indicator on a rigid reference at the measurement radius	Indicator stable, perpendicular to motion	Radius
3	Apply a small reversing force to the output side and read free movement	Repeat 3 times	Readings
4	Convert to angular backlash using the radius (OEM method)	Within the OEM limit	Value in arc-minutes
5	Compare with commissioning baseline	Increase above 50 percent triggers evaluation [I]	Ratio
6	Repeat at a second posture if permitted	Consistent	Posture

Table 21. Task T3: brake release and drag test

Step	Action	Check	Record
1	Measure brake supply voltage at the terminals during release command	Within OEM specification	Voltage
2	Observe release click or use controller log for timing	Release before motion	Delay in ms
3	With servo off and brake released under support, assess resistance of the axis by hand where the OEM allows	Free movement without drag	Subjective rating
4	Measure brake and motor temperature after a run	No local hot spot	Temperature
5	Inspect disc and armature for wear and contamination if accessible	No glazing or oil	Findings

Table 22. Task T4: fastener and coupling verification

Step	Action	Check	Record
1	Inspect torque marks on gearbox and motor fasteners	Marks aligned	Marks status
2	Apply the OEM torque with a calibrated wrench in the OEM sequence	Reaches torque without turning	Torque and tool ID
3	Replace any damaged fasteners and use thread-locking only if the OEM calls for it	-	Parts
4	Inspect coupling or key for fretting	No play	Findings
5	Apply new torque marks	-	-

Table 23. Task T5: dress pack and external drag inspection

Step	Action	Check	Record
1	Move the axis through the stroke at low speed after lock-out release under a safe procedure	Cables move freely	Posture where tight
2	Check slack at the extreme postures	Slack per OEM routing	Measurement
3	Look for rub marks and kinks	None	Photo
4	Clean spatter from covers and seals	Clean	-
5	Re-route or replace guides	-	Parts

Table 24. Task T6: Sensor 1 data-path validation

Step	Action	Check	Record
1	Open device status and diag for robot-weld-01	online; uptimeS growing; no reconnect growth	diag snapshot
2	Check seq continuity over 5 minutes	Gaps below 1 percent	Gap count
3	Check synced	true	-
4	Compare Sensor 1 with TCP and gun temperature	Plausible, not frozen	-
5	Compute sigma and the autocorrelation at 8 s	sigma 0.40 to 0.55 Nm and periodic	Values
6	If the signal is suspect, send ping and check ack; if needed reboot the gateway	ack received	ack detail

Table 25. Task T7: outline for gearbox replacement (per OEM manual only)

Step	Action	Control	Record
1	Plan: order parts and tools; schedule downtime; assign specialist	Parts verified against serial numbers	Work order
2	Lock-out; discharge energy; support forearm; disconnect motor and brake cables	Zero energy verified	Permit
3	Drain lubricant; remove motor and arm interface per OEM sequence	Parts labelled	Lubricant state
4	Inspect shafts, bearings, seals and mating surfaces	Damage assessed	Findings
5	Install new gear set or gearbox; torque in OEM sequence; fill with the OEM lubricant	Correct torque and quantity	Torque values, lubricant batch
6	Reconnect cables; remove supports; run brake and drive tests	Tests pass	Test log
7	Master axis 4 and verify TCP accuracy	Mastering within OEM	Mastering data
8	Run Post-Maintenance Testing (Section 10)	Criteria met	Baseline

Table 26. Task T8: Sensor 1 baseline recapture

Step	Action	Check	Record
1	Place robot in production, 100 percent override, tool and payload final	State valid	Program ID
2	Warm up for at least 3 cycles (simulator) or 15 to 30 minutes (real) [E]	Features stabilising	-
3	Collect at least 20 cycles	synced true, no gaps	Window
4	Compute M, template (16 phase bins), peak, minimum, sigma, A6	In healthy band	Baseline vector
5	Store with date, program and payload	-	Baseline record

11. Using the Digital Twin During Maintenance

Table 27. Twin behaviour relevant to maintenance [P]

Twin element	Behaviour	Maintenance use
Sensor binding ROBOT-WELD-01.AXIS_4_SERVO_TORQUE	Binds the channel to one mesh, suggested 105-six-axis-welding-robot/arm-swing, label Robot forearm (axis 4)	Locate the forearm on the 3D model; the mesh is a suggestion and the operator confirms the correct part
Warn state	Mesh tinted amber #d98400 steady; marker becomes a triangle	Prioritise axis 4 inspection at next stop
Alarm state	Mesh tinted red #b3261e pulsing; marker becomes a square; status message announced	Intervene
Bind flash	Mesh flashes green for about 1.6 s on successful binding	Confirm the correct mesh after rebinding
Unbound channel	Agent reports needs_binding instead of a highlight	Bind before relying on spatial pointing
Selecting a bound channel	Camera focuses on its mesh	Use for briefing the technician
Agent highlight tone	Built but unused until Phase 5, priority above alarm	Planned: highlight of the diagnosed component
Machine removed from twin	Bindings of its channels retired (kept as history)	Rebind after re-adding the machine
Telemetry retention	48 h time-series TTL	Export evidence early

12. Tooling, Spares and Competence

Table 28. Tooling and spares readiness [E]

Item	Purpose	Source of specification
Calibrated torque wrench	Fastener verification	OEM torque table
Dial indicator with stand	Backlash and play	OEM method
Contact or IR thermometer	Temperature	OEM limits
Oil sampling kit, magnet, clean containers	Lubricant analysis	Laboratory
Multimeter, clamp meter, insulation tester	Electrical tests	OEM
Lock-out and tag-out kit	Safety	Site procedure
Arm support / service fixture	Forearm support	OEM
Spare seals, lubricant, fasteners	Service	OEM parts list (not in repository)
Laptop with access to the twin and /telemetry/history	Evidence and verification	Repository API

Table 29. Competence requirements [E]

Task	Minimum competence
Visual inspection, lubricant sampling, dress pack	Trained maintenance technician
Fastener and backlash measurement	Technician with OEM training on the robot
Brake, drive and cable tests	Qualified electrician
Gearbox replacement and mastering	OEM-trained specialist
Data-path and gateway	Controls or IT engineer

13. Operator-Visible Symptoms

Table 30. Operator-visible symptoms and Sensor 1 check

Symptom seen by operator	Check in Sensor 1	Likely direction
Weld spot position drifting	dTCP and M	F01, F05, F07
Robot noisier at axis 4	A6, sigma	F01, F04
Axis 4 warm after run	Idle offset, M	F03, F02, F09
Jerky movement at one pose	Bump at one phase	F06
Clunk at direction changes	Spikes at reversals	F05
Collision stop	Spike, step in M	F07
Twin amber on axis 4	Peak 26 to 30	F01 first
Twin red on axis 4	Peak 30 or above	F01 or F07; intervene

14. Maintenance Planning by Gear-Train Stage

Table 31. Work packages by Sensor 1 stage

Stage	Sensor 1 signature	Work package	Parts to have ready	Approval and timing [I]
0 Healthy	M 12.5 to 13.5, A6 at most 0.2	Routine inspection per OEM schedule	None extra	Planned maintenance
1 Incipient	M 13.5 to 15.0, A6 0.4 to 0.6	Review lubrication records; lubricant sample; trend M and A6 each shift	Sampling kit	Supervisor informed; no stop
2 Developing	M 15.0 to 18.0, A6 0.6 to 1.5, TCP rising	Gearbox inspection package: oil, magnet, backlash, fasteners, dress pack	Seals, lubricant, fasteners; order gear set if debris found	Plan stop within days
3 Warning	Peak 26 to 30 Nm, A6 1.5 to 2.5	Full inspection; prepare replacement	Gear set or gearbox, seals, lubricant	Stop at next planned stop; specialist booked
4 Alarm	Peak 30 to 34 Nm, A6 2.5 to 3.0	Intervention; replacement	Parts on site	Within the shift; production manager informed
5 Beyond model	Sample at or above 34 Nm or M at or above 23.5	Immediate stop; damage assessment	Parts and specialist	Immediate; safety officer

15. Service Events and Their Effect on Sensor 1

Table 32. Expected Sensor 1 response to service events [E][I]

Service event	Expected short-term effect	What it means if the effect is absent or opposite	Action
Relubrication of axis 4 gearbox	M falls by 0.5 to several Nm after warm-up if friction was the cause	No change: friction was not the cause; suspect wear (A6 present?) or brake	Re-run differentiation
Lubricant over-fill	Offset rise of 0.5 to 2 Nm for a period, noise slightly up	Persisting: churning losses; correct level	Correct level and re-baseline
Replacement of the gear set	M returns to baseline within 0.5 Nm; A6 at most 0.2 Nm; run-in may add a transient	M still high: friction from new parts, brake drag, or wrong tool data	Warm-up; recheck brake and payload
Brake repair	Idle offset removed	Idle offset persists: drive offset	Drive tests
Retorque of fasteners	Reversal spikes disappear	Spikes persist: backlash or coupling wear	Backlash measurement
Dress pack adjustment	Phase bump disappears	Bump persists: other drag source	Check seals and cover contact
Tool or payload change	M and swing change, then new baseline	Ripple appears: separate wear	Recheck
Mastering	TCP back to about 0.10 mm	TCP high: backlash or kinematic data	Check parameters
Gateway reboot or reflash	Brief gap, new bootId	Torque pattern changed: not the mechanics	Check firmware version (1.0.0 in contract examples) [P]

16. Failure Prevention and Operating Practices

Table 33. Prevention measures and monitoring indicators [E]

Practice	Prevents	Indicator in Sensor 1
Warm up before production at full speed	Cold viscosity loading and early wear	Warm-up offset of M within 0.5 Nm
Verify tool and payload data after every change	Overload and wrong feedforward	M and swing stable; no step
Collision-avoidance checks when teaching programs	Gear damage	No spikes above 30 Nm
Clean spatter from covers and seals	Drag, seal damage	No slow M rise
Respect the lubrication schedule from the OEM	Lubricant starvation	M and A6 at baseline
Keep gateway clock and network healthy	Misdiagnosis	synced true; no gaps
Review the trend of M and A6 each shift	Late detection	L1 caught before L2
Reduce speed or duty at L3	Failure escalation	Peak falls below 30 Nm

17. Text Decision Trees for Troubleshooting

TREE 1 (mean high): IF M is above 13.5 Nm THEN check data path; IF data path valid THEN check state; IF state valid THEN check A6; IF A6 at least 0.4 and ratio 0.2 to 0.4 THEN go to gearbox inspection (T1, T2); ELSE check idle offset; IF idle offset at least 0.5 Nm THEN brake and drive tests (T3, electrical inspection); ELSE check temperature dependence; IF reversible THEN thermal and lubricant checks; ELSE lubricant, dress pack and tool data checks (T1, T5).

TREE 2 (spike or bump): IF an isolated spike at least 30 Nm appears once THEN check event log for collision and TCP step; IF collision suspected THEN stop and inspect; ELSE log and recheck after 5 cycles. IF spikes repeat at the same phase THEN fasteners and backlash (T4, T2); IF a smooth bump repeats THEN dress pack (T5).

TREE 3 (flat, clipped or missing): IF the value is constant or clipped THEN data-path validation (T6); IF other channels are also affected THEN gateway and network; IF only Sensor 1 is affected THEN signal scaling and controller interface; IF the program is stopped THEN treat as idle.

18. Maintenance Mistakes That Corrupt Diagnosis

Table 34. Common mistakes and consequences

Mistake	Consequence for Sensor 1 interpretation	Prevention
Recording the baseline before warm-up	M too high; later normal values look low	Warm up first
Changing tool data without re-baselining	Step in M interpreted as wear	Re-baseline after any change
Ignoring a single high peak because the mean is normal	Missing a collision	Check event log
Using per-sample status only	Missing early warning (flicker)	Use per-cycle features
Servicing without recording lubricant grade	Unexplained offset later	Record grade and batch
Closing the alert before the 600 s soak	False clearing	Follow verification
Swapping the simulator and the board without stopping one	Identity conflict	Stop one source
Treating TCP rise as proof of wear	Looseness or mastering missed	Apply ratios

19. Inspection Report Template

Table 35. Template for the maintenance record

MAINTENANCE / INSPECTION REPORT (Robot Weld 01, axis 4, Sensor 1 AXIS_4_SERVO_TORQUE)	
Date / technician / permit No.	: _____
Trigger (agent report ID, level)	: _____
Sensor 1 before work	: M _____ Nm A6 _____ Nm sigma _____ Nm peak _____ Nm TCP _____ mm
Findings: lubricant _____ magnet _____ backlash _____ fasteners _____ dress pack _____ brake _____ drive _____	
Confirmed fault (F01..F12)	: _____
Parts replaced / lubricant grade	: _____
Sensor 1 after work (20 cycles)	: M _____ A6 _____ sigma _____ peak _____ TCP _____ status normal 600 s: Y/N
Return to service approved by	: _____

20. Post-Maintenance Testing and Return to Service

20.1 Post-maintenance testing

Table 36. Post-maintenance test sequence

Test	Condition	Pass criterion	If failed
Functional test, no load	Manual low speed	No noise, no alarms	Re-inspect
Low-speed program run	25 percent override, cell clear	Smooth motion, no torque alarms	Stop
Full-speed warm-up	3 cycles (simulator) or 15 to 30 minutes (real) [E]	Features stabilise	Continue to warm-up
Baseline capture	20 cycles	M 12.5 to 13.5 Nm; A6 at most 0.2; sigma 0.40 to 0.55; peak at most 23.1	Reopen
TCP check	20 cycles	TCP mean at most 0.15 mm	Check mastering
Soak	600 s	No warn or alarm	Reopen
Record	-	Baseline stored	-

20.2 Return-to-service criteria

Table 37. Return-to-service criteria

Criterion	Requirement	Evidence
Safety	Guards in place, lock-out removed, cell clear	Permit closed
Mechanical	Fasteners torqued, backlash within OEM, no leaks	Checklist
Electrical	Brake and drive tests passed	Logs
Telemetry	status online, synced true, no gaps	diag snapshot
Sensor 1 features	Healthy band over 20 cycles	Document 03 criteria
Platform status	normal for 600 s	Twin view
Documentation	Findings, parts and baseline recorded	Maintenance record
Monitoring	Elevated watch for the next 24 h	Agent setting

21. Escalation Criteria

Table 38. Escalation criteria for maintenance

Condition	Escalate to	When
Collision or L4	Specialist and safety officer	Immediately
Alarm persists after corrective action	Specialist	Immediately
Metal debris found	Gearbox specialist	Same shift
Brake or drive fault	Electrician and OEM service	Same shift
Unresolved diagnosis after two inspections	Specialist	Next stop
Data path fault persisting 24 h	Controls engineer	Next day
Any unsafe condition	Safety officer	Immediately

22. Data-Path Troubleshooting for Sensor 1

An implausible or missing Sensor 1 reading must be tested against the data path before any mechanical work. The table reproduces the repository's documented symptoms [P].

Table 39. Data-path symptoms and causes [P]

Symptom	Likely cause	Source
Telemetry arrives but the twin shows nothing	Channel not bound to a mesh, or machine not added to the twin	firmware README, cheat sheet
Device shown stale	No telemetry for max(5 s, 4 x intervalMs)	MQTT contract
Device offline about 15 s after power loss	Last Will published by the broker	firmware README
Identity conflict flagged	Board and simulator both announcing robot-weld-01	demo script
A fault seems stuck	Choose Normal in the Faults tab or send cmd/reboot	demo script
No Faults tab	ENABLE_DEVICE_COMMANDS is not true	demo script
wifi disconnect event, reason 201	Network not found, or 5 GHz only	firmware README
reason 202 or 15	Wrong Wi-Fi password or unsupported security mode	firmware README
mqtt connect failed (state 4) or state 5	Wrong broker credentials, or credential lacks permission	firmware README
tls error -9984	Clock wrong, or CA file does not anchor the chain	firmware README
clock not set after 15 s	Network blocks NTP (UDP 123)	firmware README
Gaps in seq	Packet loss (a new bootId with seq 0 is a reboot, not loss)	MQTT contract

23. Maintenance Records and Feedback to the Diagnostic System

- Record per intervention: date, machine, axis, Sensor 1 level and features before work, findings, parts, lubricant grade, torque values, backlash, post-work features, technician.
- Feed findings back: label the confirmed fault (F01 to F12) so that future cases can be added to Document 05 as real history. At present no plant history exists in the repository [P].
- Telemetry TTL is 48 h in the platform; export evidence before it expires if it must be kept [P].

24. Glossary of Terms Used in the Maintenance Documents

Table 40. Glossary

Term	Meaning in this suite
Sensor 1	AXIS_4_SERVO_TORQUE, axis 4 servo torque in Nm, sensor ID ROBOT-WELD-01.AXIS_4_SERVO_TORQUE
Axis 4	The forearm axis of the six-axis welding robot
Cycle	The 8 s weld cycle; gun energised 4.4 to 6.0 s
M	Mean torque over a window of whole cycles (Nm); healthy 13.0
S	Severity index $(M - 13.0) / 10.5$, equivalent to the simulator fault ramp r
A6	Amplitude of the sixth harmonic of the cycle (0.75 Hz) in Nm
sigma	Noise standard deviation after removing the cycle template (Nm); healthy 0.45
TCP deviation	TOOL_CENTER_POINT_DEVIATION in mm (Sensor 2 channel), used here only for corroboration
GEARBOX WEAR	The only robot fault scenario in the repository: damaged gear mesh in axis 4
Watch / Warning / Alarm / Critical	L1 / L2 / L3 / L4 levels defined in Document 01
warnHigh / alarmHigh	Limits 26 and 30 Nm declared by the machine
LOTO	Lock-out and tag-out
OEM	Original equipment manufacturer
needs_binding	Agent output when the channel is not bound to a mesh on the twin

25. Summary, Safety Notes, References and Revision

25.1 Maintenance summary

- Maintenance is condition-based: Sensor 1 levels and gear-train stages decide what to do; OEM intervals apply to routine work.
- First checks for wear suspicion: oil and magnet, backlash, fasteners, dress pack; first checks for friction suspicion: lubricant, brake, temperature; first checks for data suspicion: diag, seq, synced.
- Key numbers: warn 26 Nm, alarm 30 Nm, critical 34 Nm (proposed); healthy M 12.5 to 13.5 Nm; A6 at most 0.2 Nm; TCP at most 0.15 mm; 600 s soak.

25.2 Safety and escalation notes

Lock-out and tag-out, OEM procedure and site safety rules apply to every task. Support the forearm before brake release. This document is a capstone training stub, not a replacement for the OEM manual.

25.3 References and engineering basis

- Repository: runbook stubs for the stamping press (pattern and safety wording), firmware README and MQTT contract (data-path), Phase 5 blueprint.
- General industrial robot maintenance practice and ISO 10218 safety context [E]; all OEM-specific values to be taken from the robot manual.

25.4 Revision notes

Table 41. Revision history

Rev	Date	Change
1.0	2026-10-08	Initial release.